The Influence of Time of Season on Injury Rates and the Epidemiology of Canadian Football Injuries

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Abstract

Objective: To describe injury rates and injury patterns in the Canadian Football League (CFL) according to time during the season, player position, injury type, and injury location. **Design:** Prospective, cohort study. **Setting and Participants:** Eight seasons from CFL injury surveillance database. **Independent Variables:** Depending on the analysis, time of season (preseason, regular, and playoffs), player position, injury type, and injury location. **Main Outcome Measures:** Medical attention and time-loss injury rates per 100 athletes at risk (AAR), and prevalence of time-loss injuries per week. **Results:** The average game injury rate was 45.2/100 AAR medical attention injuries and 30.7/100 AAR time-loss injuries. Injury rates declined by 1% per week over the season for both medical attention (rate ratio = 0.99) and time-loss (rate ratio = 0.99) injuries, with a substantial decline during the playoffs compared with preseason (rate ratio = 0.70-0.77). The number of ongoing time-loss injuries increased over the course of the regular season. Quarterbacks, offensive backs, and linebackers had the highest game injury rates. Joint/ligament and muscle/tendon injuries were the most common injury types for games and practices, respectively. The lower extremity was the most commonly affected area, specifically the lower leg/ankle/foot and hip/groin/thigh. **Conclusions:** There was a 1% decline in injury rate per week during the season and a 30% decline during the playoffs. The number of ongoing time-loss injuries increased over the regular season. Current results can aid league officials and medical staff in making evidence-based decisions concerning player safety and health.

Key Words: Canadian Football League, football, injury rate

(*Clin J Sport Med* 2020;00:1–7)

INTRODUCTION

Throughout their career, football players are at risk of sustaining injuries due to the inherent nature of the sport. Football has the highest injury rate in games and practices when compared with 14 other National Collegiate Athletic Association (NCAA) sports.[1] The time-loss injury rate for NCAA football games is between 11.6 and 40.2/1000 athlete exposures (AE), which represents a substantial burden on medical teams.[2–5]

Submitted for publication August 14, 2019; accepted December 4, 2019.

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S. M. Robbins is supported by the Fonds de recherche du Québec–Santé [grant number 33107]. Sources of funding had no role in this project. D. Naidu acts in an unfunded position as research chair of the Canadian Football League and is the head team physician of the Edmonton Eskimos.

The authors report no conflicts of interest.

Data from this study are not publically available.

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http://dx.doi.org/10.1097/JSM.0000000000000824

Football injury rates change throughout the season.[3] Injury data from over 100 NCAA football programs demonstrated that the time-loss game injury rate during the regular season (36.11/1000 AE) was higher than preseason (7.55/1000 AE) and playoffs (23.71/1000 AE).[2] By contrast, the practice time-loss injury rate during preseason (7.24/1000 AE) was higher than regular season (2.09/1000 AE) and playoffs (1.35/1000 AE).[2] These findings are supported by another study that examined 60 NCAA football programs.[5] In a study of one National Football League (NFL) team, time-loss injury rates were higher in the first 2 weeks compared with the last 3 weeks of training camp.[6] Although these results generally suggest that injury rates decline over the preseason and regular season, there are limited data on professional football teams. In addition, the above results refer only to new injuries, and there is little information in regards to ongoing injuries.

Player position affects injury rate and the frequency of injury types, and anatomical locations vary.[7] In the NCAA, running backs sustained the highest proportion (19.6%), and offensive lineman sustained the lowest proportion (9.9%) of injuries.[2] The most common game and practice injury types in NFL preseason are joint sprains and muscle strains, respectively.[6] In terms of location, the lower extremity (54.7%) accounted for the greatest percentage of game injuries, with the knee and ankle being most often injured.[2] Further data on professional players are not available, where differences in age, size, and skill could affect injury patterns.

The professional football league in Canada, the Canadian Football League (CFL), represents an unstudied context.

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Changes in weekly injury rates throughout the season have not been investigated in this league. Canadian Football League competition level (professional) differs from the NCAA (amateur), and CFL season length differs from both NCAA and NFL. Furthermore, differences in game rules could impact the style of play including fewer downs (3 vs 4 downs), more players on the field (12 vs 11), and larger fields (110 by 65 vs 100 by 53.3 yards). These differences might affect injury rates, player position most at injury risk, injury types, and injury locations. Therefore, our primary objective was to describe variations in CFL injury rates over a season using 8 years of play. We hypothesized that injury rate during the preseason would be high with a drop during the early regular season and plateau as the season progressed. Secondary objectives were to describe the burden of injury (prevalence of ongoing time-loss injuries) over the season, as well as injury patterns according to player position, injury type, and injury location.

METHODS

Study Population and Data Collection

We used routinely collected data from the CFL injury surveillance database, which serves as the official electronic record for health care professionals from each team. Data were prospectively entered by team athletic therapists, physical therapists, and medical doctors. There were 8 teams for 2011 to 2013 seasons and 9 for 2014 to 2018 seasons. Game and practice injury data were extracted including player position, anatomical location, injury date, diagnosis according to the Sport Medicine Diagnostic Coding System,[8] and medical clearance date. All CFL registered players in 2011 to 2018 were included.

Injury Definitions

We included nontraumatic and traumatic musculoskeletal and neurological injuries that occurred during games or practices during the season. We excluded injuries and illness categorized as cardiovascular, respiratory, dermatology, endocrine, environmental, fluid and electrolyte, gastrointestinal, genitourinary, hematology, infectious disease, psychology, metabolic bone, and unassigned. We also excluded blisters, abrasions, and lacerations, which are often included in injury studies but have limited consequences and only accounted for 1.34% of all injuries. Non–football-related injuries, injuries occurring in the weight room, offseason injuries, and preventative medical appointments were excluded. Analyses were conducted for both medical attention and time-loss injuries. Medical attention injuries included those that occurred during games or practices, and injured players sought assessment/treatment from team medical staff. We considered injuries to result in time loss if the medical clearance date was not the same as the first visit date. We also examined injury burden (ie, ongoing injuries) over time, which we defined as the number of players with a time-loss injury that was not medically cleared to play for one or more days during a given week.

Weekly Injury Rate

Injury rate corresponds to the number of injuries divided by a unit of person time.[9] We used athletes at risk (AAR) expressed as injuries per 100 AAR for our denominator of risk because the more commonly used AE method includes many athletes in the denominator who are not at risk of injury.[9] Canadian Football League has 12 players per team on the field at any point in a game who are at risk of game injuries during active play, except under exceptional circumstances. The total number of AAR in one week is 12 multiplied by the number of teams that played on a specific week, and the injury rate is equal to the number of injuries/AAR (expressed per 100 AAR, similar to per 1000 AE). Injury rates were calculated for both medical attention and time-loss injuries that occurred during games. For practices, we present total number of injuries because we did not have data on the number of players practicing each week.

Period

For CFL seasons occurring from 2011 to 2013, there were 3 weeks of preseason and 19 weeks of regular season (18 regular season games and 2 preseason games with 1 bye week where some teams had no games). For 2014 to 2017, there were 3 weeks of preseason and 20 weeks of regular season (18 regular season games and 2 preseason games with 2 bye weeks). In 2018, there were 3 weeks of preseason and 21 weeks of regular season (18 regular season games and 2 preseason games with 3 bye weeks). There were 3 weeks of playoffs for all seasons, where teams could play 0, 1, 2, or 3 games.

To determine weekly injury rates, weeks were categorized according to the CFL website.[10] Preseason weeks were categorized as follows: 0 for the last preseason week before the beginning of the regular season, −1 for the second to last week before regular season, and −2 for the third to last week. The first week of the regular system was categorized as 1, with subsequent regular and playoff weeks increasing by one number each week. Weeks were also coded according to whether they were preseason, regular season, or playoffs. Because season lengths differ, weeks 20 and 21 were sometimes regular season games and sometimes playoff games.

Player Positions

Player positions were categorized into offense, defense, and special teams. Offensive players were further categorized into more specific positions: quarterback, lineman (center, guard, and tackle), offensive backs (running back and full back), and receivers (wide receiver, slot back, and tight end). Defensive players were categorized into linemen (end and tackle), linebackers, and defensive backs (cornerback and safety). The positions kicker/punter, long snapper, and kick returner were categorized as special teams.

For game injury rates across offensive and defensive player positions, we calculated AAR using the number of players in each position during usual formations based on CFL depth charts and discussions with CFL personnel. Results refer to the risk of injury in each player of that position to compare injury risk of playing as a quarterback to playing as a lineman, although there are several linemen on the field at one time. The offensive formation included 1 quarterback, 5 offensive lineman, 2 offensive backs, and 4 receivers. The defensive formation included 4 defensive linemen, 3 linebackers, and 5 defensive backs. Team

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formations will vary throughout a game, but they represent the most common formations. For practices, we report only the frequency of injuries (overall count) by player position, as the number of players at each practice was not recorded.

Injury Types

Injuries were classified according to their type based on a nominal consensus model approach for soccer injuries (no consensus statement on American or Canadian football injuries has been reached to date).[11] Injury-type groupings consisted of fractures, joint/ligament, muscle/tendon, contusions, central/peripheral nervous system, and other.

Anatomical Location

Consistent with previous studies, the anatomical location of injury was classified as head and neck; shoulder; upper extremity (upper arm, elbow, forearm, hand, and wrist); trunk (abdomen, chest/ribs/upper back, and pelvis/low back); hip, groin, and thigh; knee; lower leg, ankle, and foot; and nonspecific musculoskeletal injuries.[4,7,12]

Analysis

We compared injury rates across weeks using Poisson regression, with weekly injury rate as the outcome, "CFL week" as the main exposure variable, time of season (preseason, regular season, and playoffs) as a covariate (reference = preseason), and AAR as an offset to account for the different number of teams playing in any given week. We report rate ratios with 95% confidence intervals (95% CIs). Separate regression analyses were conducted for medical attention and time-loss game injuries. In addition, a sensitivity analysis was conducted for time-loss injuries where only injuries that resulted in players missing 8 or more days were considered as time-loss injuries. Missing 8 or more days would likely indicate that a player missed at least one game. Underlying assumptions of the models were verified.

Injury burden was operationalized as the number of players with a time-loss injury that was not medically cleared to play for at least 1 day during a CFL week. We present the mean injury burden per CFL week across the 8 years. Injuries by player position were reported as injury rates for game injuries and total counts for practice injuries. Proportions of injuries were determined based on injury types and anatomical locations. Statistical analyses were completed with R version 3.5.1.[13]

Ethical Considerations

This study was approved by the Institutional Review Board of McGill University Faculty of Medicine.

RESULTS

Original data included 15 618 injuries and illnesses. After applying inclusion/exclusion criteria, medical attention injuries included 7914 game and 3739 practice injuries, and time-loss injuries included 5372 game and 2204 practice injuries (Figure 1). There were 69 preseason games, 621 regular season games, and 40 playoff games over 8 seasons. The average medical attention game injury rate was 45.2/100 AAR (95%

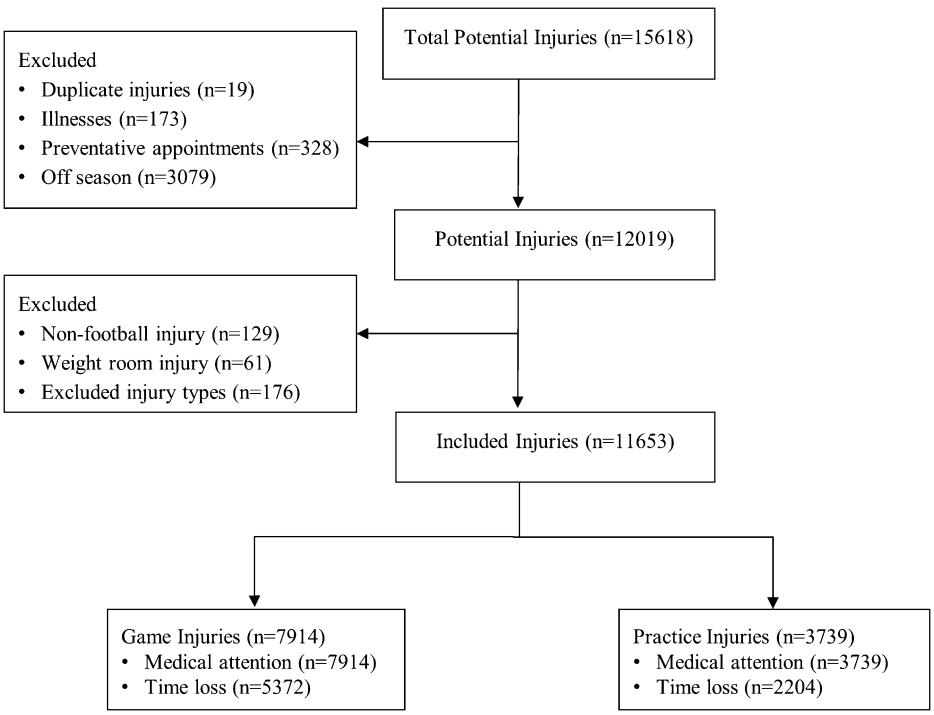

Figure 1. Flow diagram showing the exclusion criteria applied to the data and the final number of game and practice injuries.

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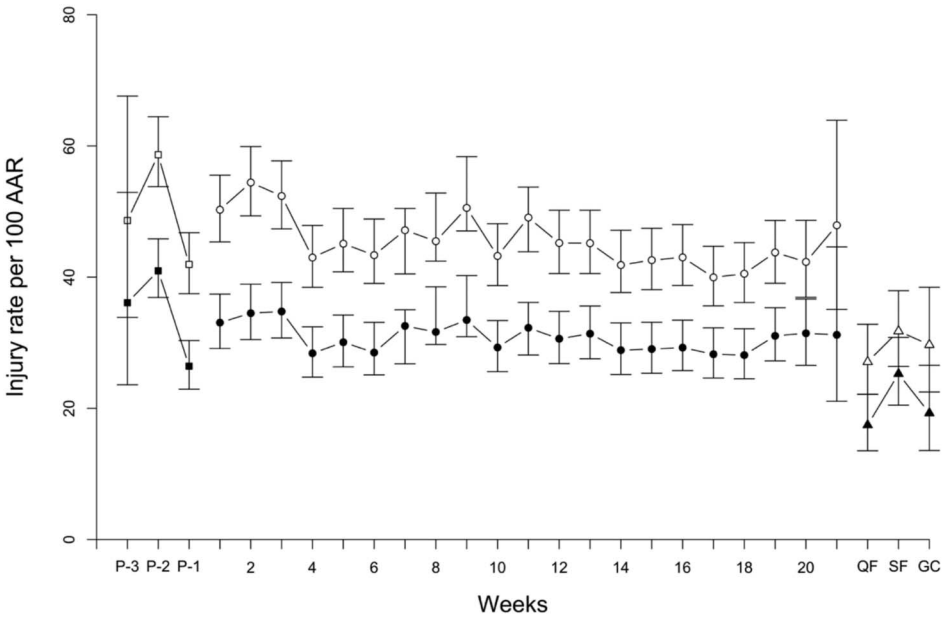

Figure 2. The average weekly injury rate for medical attention injuries (white) and time-loss injuries (black) across the 8 seasons for the preseason (squares), regular season (circles), and playoffs (triangle). Error bars represent 95% CIs. The last preseason game before the regular season is P-1. The second and third to last preseason games are coded as P-2 and P-3, respectively. Playoffs are coded as quarterfinals (QF), semifinals (SF), and Grey Cup final (GC). The length of the season varied over the 8 years, and for this figure, the playoff weeks were not continuous but aligned based on playoff game (QF, SF, and GC). Weeks were treated as continuous in the Poisson regression with additional variables accounting for game classification (preseason, regular season, and playoffs).

CI: 44.4-45.9), and time-loss game injury rate was 30.7/100 AAR (95% CI: 30.0-31.4).

Weekly Injury Rate

For medical attention injuries, there was a meaningful decrease in the weekly injury rate from preseason to playoffs (rate ratio = 0.73; 95% CI: 0.62-0.86), but not from preseason to regular season (rate ratio = 1.01; 95% CI: 0.93-1.10) (Figure 2). There was also a meaningful decrease in the weekly injury rate across the entire season of 1% per week, which represents a 22% decrease over a 20-game regular season (rate ratio = 0.99; 95% CI: 0.99-0.99).

Similar to medical attention injuries, time-loss weekly injury rates decreased from preseason to playoffs (rate ratio = 0.70; 95% CI: 0.58-0.85), but not from preseason to regular season (rate ratio = 0.97; 95% CI: 0.87-1.07). Time-loss injuries also decreased by 1% per week across the entire

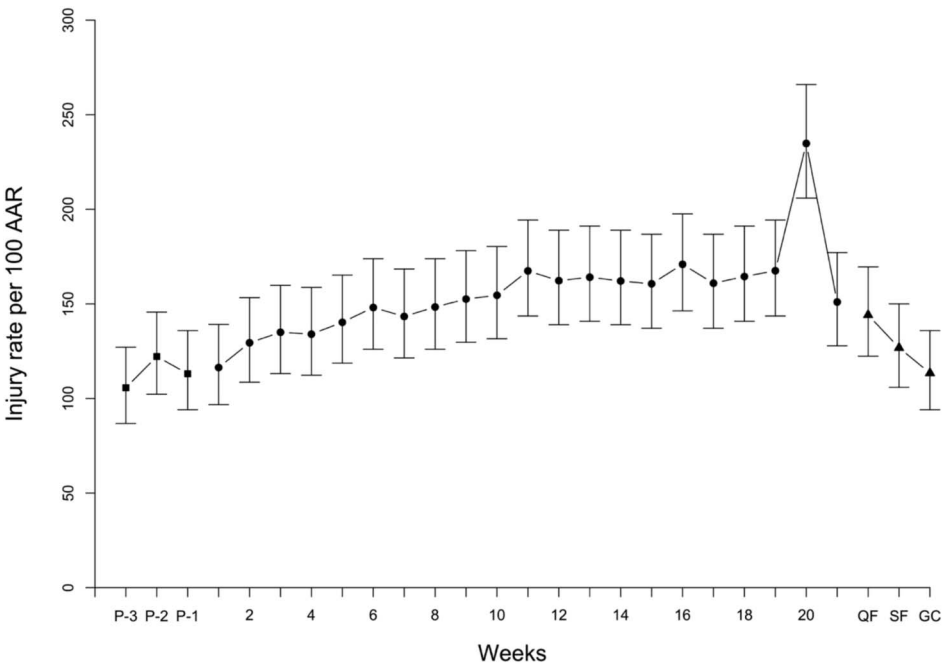

Figure 3. The injury burden representing the mean number of ongoing time-loss injuries each week across the 8 seasons for the preseason (squares), regular season (circles), and playoffs (triangle). Error bars represent 95% CIs. The last preseason game before the regular season is P-1. The second and third to last preseason games are coded as P-2 and P-3, respectively. Playoffs are coded as quarterfinals (QF), semifinals (SF), and Grey Cup final (GC).

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TABLE 1. Number of Medical Attention and Time-Loss Injuries, and Injury Rate According to Player Positions

Player Position	Game				Practice	
	Medical Attention Total Injuries	Medical Attention Injury Rate (/100 AAR)	Time-Loss Total Injuries	Time-Loss Injury Rate (/100 AAR)	Medical Attention Total Injuries	Time-Loss Total Injuries
Offense (total)	3519	40.2	2443	27.9	1615	930
Offensive linemen	1047	28.7	732	20.1	534	306
Quarterback	427	58.5	245	33.6	122	59
Offensive back	782	53.6	566	38.8	289	180
Receiver	1263	43.3	900	30.8	670	385
Defense (total)	4039	46.1	2686	30.7	1700	1023
Defensive linemen	1218	41.7	813	27.8	550	331
Linebacker	1194	54.5	794	36.3	415	245
Defensive back	1627	44.6	1079	29.6	735	447
Special teams	135		79		85	40
Kicker-punter	90	—	53	—	69	32
Long snapper	37	—	18	—	10	2
Kick returner	8	—	8	—	6	6
Missing information*	221	—	164	—	339	211

** Player position data were not available for players who only played in the preseason and were cut from the roster before the regular season.*
AAR: at any one point in time during a game, the number of AAR for injury is the number of players on the field (12 per team).

season (rate ratio = 0.99; 95% CI: 0.99-1.00). The rate ratios were similar when we restricted injuries to those missing 8 or more days [playoffs to preseason = 0.77 (95% CI: 0.59-1.00); preseason to regular season = 0.92 (95% CI: 0.81-1.06); and weekly decline across the entire season = 0.99 (95% CI: 0.99-1.00)].

Injury burden, or the number of ongoing time-loss injuries, in a week increased throughout the regular season and then declined during the playoffs (Figure 3).

Player Positions

Defensive players had higher game injury rates for medical attention and time-loss injuries compared with offensive players. For specific positions, quarterbacks had the highest medical attention game injury rate and offensive backs had the highest time-loss game injury rate. Linebackers had the highest game injury rates for defensive players (Table 1).

When examining practice injuries, defensive players had higher numbers of medical attention and time-loss injuries; defensive backs sustained the most practice injuries of any specific position (Table 1). Because we did not have details on which position athletes attended practices, these numbers represent the total number of injuries for a position. Therefore, if a team has 10 offensive linemen and 2 quarterbacks practicing, one would expect 5 times the number of injuries if the injury risk was the same across positions.

Injury Types

For both medical attention and time-loss injuries during games, joint/ligament injuries were the most common type, followed by a relatively equal number of injuries classified as muscle/tendon and contusions. For practice injuries, muscle/tendon was the most common injury type followed by joint/ligament for both medical attention and time-loss injuries (Table 2).

TABLE 2. The Number and Proportion of Each Injury Type for Both Game and Practice Injuries

Injury Type Categories	Game		Practice	
	Medical Attention [% (n)]	Time Loss [% (n)]	Medical Attention [% (n)]	Time Loss [% (n)]
Joint/ligament	33 (2620)	35 (1893)	26 (954)	26 (584)
Muscle/tendon	23 (1788)	24 (1288)	40 (1502)	43 (946)
Contusions	21 (1696)	17 (903)	13 (486)	11 (238)
Central/peripheral nervous system	7 (581)	9 (468)	3 (100)	4 (91)
Fracture	4 (288)	5 (242)	3 (102)	3 (70)
Other	12 (941)	11 (578)	16 (595)	12 (274)

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TABLE 3. The Number and Proportion of Injuries at Each Anatomical Location for Both Game and Practice Injuries

	Game		Practice	
Anatomical Location	Medical Attention [% (n)]	Time Loss [% (n)]	Medical Attention [% (n)]	Time Loss [% (n)]
Lower leg, ankle, and foot	22 (1771)	23 (1248)	27 (993)	26 (569)
Hip, groin, and thigh	16 (1276)	16 (884)	25 (945)	26 (568)
Upper extremity	16 (1255)	13 (702)	11 (414)	10 (214)
Knee	15 (1225)	17 (928)	15 (567)	16 (355)
Head and neck	11 (895)	12 (625)	6 (216)	6 (138)
Shoulder	11 (979)	11 (609)	6 (233)	7 (148)
Trunk	8 (605)	7 (371)	10 (366)	9 (209)
Nonspecific/missing	0 (8)	0 (5)	0 (5)	0 (3)

Anatomical Location

Locations most commonly injured during games and practices resulting in a medical attention injury included the lower leg, ankle, and foot; hip, groin, and thigh; upper extremity; and knee. A similar trend was found for time-loss injuries (Table 3).

DISCUSSION

Few studies have examined injury rates in professional football, and this is the first to examine the CFL. Medical attention and time-loss injury rates over 8 CFL seasons were 45.2/100 AAR and 30.7/100 AAR, respectively. Injury rate per week declined over the regular season, with a substantial decline during the playoffs. Although the number of new injuries declined during the season, there was an increase in the number of ongoing injuries as the season progressed (ie, injury burden), although a decline again occurred during the playoffs. Medical staff can use this information to plan care and allocate resources to address new and ongoing injuries.

The CFL time-loss game injury rate (30.7/100 AAR) was lower than in previous NFL studies. In one NFL study examining preseason games for one team over 10 seasons, the injury rate was 53.0/100 AAR (converted from 64.7/1000 AE).[6] Using online NFL injury reports, another study found the combined game and practice injury rates were 39.6/100 AAR (reported as 395.8/1000 AAR in the original publication).[14] Differences in injury definitions, rate calculations, and player size could account for these discrepancies.

The decline in game injury rates throughout the season, with a substantial drop off during the playoffs, is consistent with NCAA studies.[2,3,5] One possible explanation for the decrease during the playoffs is that players may report less injuries since the season is ending, and they feel they can manage injuries in the offseason. Alternatively, they may not report injuries due to fear of not being allowed to play during playoffs. Both hypotheses are speculative, and further exploration into reasons why these changes occur should include an analysis of player training patterns, beliefs, and application of medical care.

Injury burden (ie, ongoing time-loss injuries) increased during the regular season. This means more injuries occurred each week than were healed each week. From a management perspective, this increases workload on medical staff near the end of the season when playoffs are approaching. Some teams may feel it helpful to hire additional staff later in the season. However, there was a sharp decline during the playoffs. Because fewer teams are playing during the playoffs, players on eliminated teams are likely to heal much faster, and there are fewer AAR of incurring new injuries.

The most commonly injured positions were quarterbacks, offensive backs, and linebackers. This is similar to a previous NCAA study that calculated time-loss injury rates by position using similar methods.[2] This finding is not surprising for running backs and linebackers since their collisions typically occur at higher speeds than other positions with the force in the opposite direction of their motion. For quarterbacks, they may be left exposed during passing attempts.

In regards to injury types, joint/ligament injuries were the most common game injury, and muscle/tendon injuries were the most common practice injury. A similar pattern was found in the NFL preseason.[6] High proportions of joint/ligament injuries during games are not surprising given the increased number and speed of collisions and increased unpredictability. High proportions of muscle/tendon injuries during practice could represent overuse injuries[6] and potential training errors (eg, large increase in training volume during the preseason) could be addressed preemptively.

In terms of injury location, lower extremity was the most commonly injured area. Similar findings have been reported in other leagues.[2,12,14] These findings that address injury proportions based on position, type, and location can aid league administrators in making evidence-based decisions to improve current programs on player safety and medical staff on planning the appropriate care.

Our study has some limitations. First, data accuracy depended on information reported by players and medical staff. Underreporting of injuries by an athlete is a concern in every injury study and is difficult to compensate for, as injury rates can only be based on verifiable facts. Second, practice injury rates could not be calculated because we did not have data on which athletes attended practices. Finally, similar to other studies on football, our injury rate calculations considered 2 injuries to 2 different athletes the same as 2 injuries to the same athlete.

In conclusion, there was a slight decline in CFL injury rates over the season, with a substantial decline in the playoffs. The injury burden increased during the regular season but declined during the playoffs. These findings underscore the need for continued proactive injury monitoring with appropriate offseason care. Quarterbacks, offensive backs, and

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linebackers had the highest injury rates. Joint/ligament and muscle/tendon injuries were the most common injury types with the lower extremity the most commonly affected area. Such information can aid league officials and medical staff in making evidence-based decisions concerning player safety and health.

ACKNOWLEDGMENTS

The authors thank Steve Daniel (CFL Director, Game Information and Statistics), Kevin McDonald (Vice President of Health and Safety, CFL), and the Canadian Football League for providing assistance.

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